

Hydrologic Response Units in OpenHydroQual

The HRU and Urban HRU composite blocks

Abstract

A hydrologic response unit (HRU) in OpenHydroQual is a composite block that bundles the surface, soil and groundwater stores of one piece of land and the connectors between them, so that a sub-catchment is added to a model as a single object with a single set of properties. This document describes the two HRUs: the `Hydrologic_Response_Unit` (“HRU”), a single infiltrating surface over a three-layer soil column and a groundwater cell, and the `Urban_HRU`, which divides the land into pervious and impervious surfaces with their own routing reaches, places the groundwater cell under the whole unit as an unconfined aquifer whose initial state is a water-table elevation, recharges it through the unsaturated conductivity of the bottom soil layer, can draw evapotranspiration from the deeper soil layers through a second source, takes soil parameters layer by layer and carries three calibration multipliers. For each unit the document sets out the members and connectors, the equation of each flux, how the plan area is divided, how the vertical geometry and the initial state follow from a few properties, and how the impervious fraction should be estimated, and it closes with a worked example and a property reference.

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1 Which HRU to use

	Hydrologic_Response_Unit	Urban_HRU
File	hydrologic_response_unit.json	urban_hru.json
Surfaces	one, infiltrating	pervious + impervious
Internal routing reaches	none	one per surface
Soil parameters	one set	one set per layer
Groundwater area	A	A
Recharge conductivity	constant <code>recharge_K</code>	$K_{sat,3} k_r(S_{e,3})$
Groundwater cell	<code>Groundwater_cell</code> , user S_s	<code>Unconfined Groundwater_cell</code>
Groundwater template	<code>groundwater.json</code>	<code>unconfined_groundwater.json</code>
Initial groundwater state	moisture content	water-table elevation
Evapotranspiration	<code>Soil_1</code>	<code>Soil_1</code> ; optional second source on <code>Soil_2–Soil_3</code>
Calibration multipliers	—	<code>impervious_multiplier</code> , <code>K_sat_scale</code> , <code>K_sat_scale_surface</code>
Icon		

Table 1: The two HRU composites. A is the plan area, ϕ the groundwater porosity and b the groundwater thickness.

Use the **HRU** for land without significant impervious cover — forest, fields, parks, hillslopes — and for cascades of units where the overland flow of one unit runs onto the next (Section 3). Use the **Urban HRU** wherever part of the land is impervious and drains directly to a stream or storm drain: it partitions the area, routes each surface separately, and its subsurface (Section 4) stays physically sensible when the water table lies well below the soil column.

2 Elements shared by both units

Both units build their subsurface from the same members: three `Soil` layers and a `Groundwater cell`, connected by infiltration, two percolation links and a recharge link. The geometry and the soil-water relations below apply to both; the units differ in the plan areas they give these members, the recharge link and the groundwater storage (Sections 3 and 4).

2.1 Vertical geometry

The user gives three levels; everything else is derived:

Member	Thickness	Bottom elevation
Catchment, <code>Impervious_Catchment</code> , reaches	—	z_s (<code>surface_elevation</code>)
<code>Soil_1</code>	$D/7$	$z_s - D/7$
<code>Soil_2</code>	$2D/7$	$z_s - 3D/7$
<code>Soil_3</code>	$4D/7$	$z_s - D$
<code>Groundwater</code>	b (<code>groundwater_thickness</code>)	$z_s - D - b$

D is `depth_to_groundwater`. The 1 : 2 : 4 split puts the thinnest layer at the surface, where moisture gradients are steepest. The groundwater cell sits directly below the soil column.

2.2 Soil water

Each soil layer stores water $V = \theta A_\ell d_\ell$ and has the van Genuchten–Mualem relations

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r}, \quad \psi = -\frac{1}{\alpha} (S_e^{-1/m} - 1)^{1/n}, \quad k_r = \left[1 - (1 - S_e^{1/m})^m \right]^2, \quad m = 1 - \frac{1}{n}, \quad (1)$$

with S_e bounded to $[0.001, 0.999]$, the suction $|\psi|$ capped at 10^4 m, and the total head $h = z_{\text{bottom}} + d_\ell + \psi$ (plus a small excess pressure above saturation, governed by the specific storage). The layer's conductivity is $K_{\text{sat}} = \text{K_sat_original} \times \text{K_sat_scale_factor}$.

Infiltration

(surfacewater_to_soil_link, surface $\rightarrow$ Soil_1): $Q = \min(A_s, A_1) K_{\text{sat},1} (h_s - h_1) / (d_1/2) \cdot d / (d + 0.001)$, i.e. Darcy flow over half the top layer, switched off smoothly when no water is ponded.

Percolation

(soil_to_soil_link): $Q = \bar{A} K_{\text{eff}} (h_i - h_j) / \bar{d}$ with $K_{\text{eff}} = \min(K_{\text{sat},i}, K_{\text{sat},j}) k_r(\max(S_{e,i}, S_{e,j}))$, mean area $\bar{A}$ and mean thickness $\bar{d}$. Flow can be upward (capillary rise).

Evapotranspiration

is withdrawn from Soil_1 by the source connected to the Evapotranspiration property. With the Evapotranspiration_Time_Series (Soil) source the rate is $\text{corr_fact} \times ET_0 \times A_1 \theta / \theta_s$. The Urban HRU can also withdraw from Soil_2 and Soil_3 through a second source (Section 4.9).

3 HRU

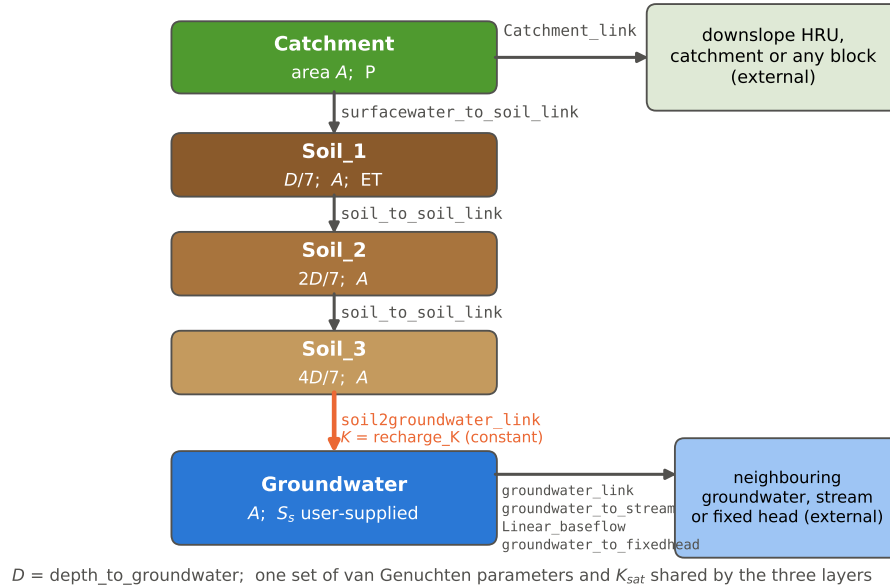


Figure 1: Hydrologic_Response_Unit: members, internal links and ports.

The HRU (hydrologic_response_unit.json, Fig. 1) has five members — a Catchment and the shared soil column and groundwater cell — and four internal links. All of them receive the

whole plan area A . It has no internal reaches: the surface drains through its `Catchment_link` port directly to whatever block it is linked to, which makes cascades of units possible (the overland flow of an upslope unit runs onto the surface of the next). Its ports:

Port (link type)	From member	Typical target	Icon
<code>Catchment_link</code>	<code>Catchment</code>	a downslope HRU, a catchment, a channel or any block	
<code>groundwater_link</code>	<code>Groundwater</code>	the groundwater cell of a neighbouring unit	
<code>groundwater_to_stream</code>	<code>Groundwater</code>	a channel segment (baseflow, head-driven)	
<code>Linear_baseflow</code>	<code>Groundwater</code>	a channel segment (baseflow, linear reservoir)	
<code>groundwater_to_fixedhead</code>	<code>Groundwater</code>	a <code>fixed_head</code> block	

Baseflow can leave the groundwater cell through either of two connectors (use one of them for a given discharge pathway, not both in parallel):

`groundwater_to_stream`

Darcy flow driven by the head difference between the groundwater cell and the receiving segment, $Q = K A_b (h_g - h_c)/L$, with the connector's exchange area A_b , flow path L and conductivity K . Discharge stops, and reverses, when the stream stage rises above the water table.

`Linear_baseflow`

A linear reservoir above an inactive storage (`groundwater.json`):

$$Q = k (\theta_g - \theta_{\min})_+ V_g^{\text{tot}}, \quad (2)$$

where θ_g is the moisture content and V_g^{tot} the total volume of the groundwater cell, k the `recession_rate` (1/d) and $\theta_{\min}$ the `min_moisture_content` below which no baseflow leaves (an absolute moisture content, not larger than the porosity). With no recharge the active storage recedes exponentially with time constant $1/k$. The discharge does not depend on the stream stage or on `groundwater_K`; k is set, or calibrated, directly from observed recession rates. An example is in `Examples/Linear_Baseflow/Linear_Baseflow.ohq`.

Recharge uses the stock `soil2groundwater_link` with the constant conductivity `recharge_K`,

$$Q_r = A K_{rech} \frac{h_3 - h_g}{L}, \quad L = \frac{1}{2} (d_3 + b), \quad (3)$$

and the groundwater member is the stock `Groundwater cell`, whose head is $h_g = z_g + b + (\theta_g - \phi)/S_s$ with user-supplied porosity ϕ and specific storage S_s , and whose initial state is a moisture content $\theta_{g,0}$. With the defaults ($S_s = 0.01 \text{ m}^{-1}$) the cell responds as a *confined* aquifer: a change of 0.01 in moisture content moves the head by 1 m; $S_s = \phi/b$ gives unconfined behaviour, and $\theta_{g,0} = \phi + (h_0 - z_g - b) S_s$ starts the water table at h_0 . A constant recharge conductivity is adequate when the water table lies at or just below the soil column; when it lies far below, see Section 4.6. One set of soil parameters applies to all three layers. A two-unit hillslope using the HRU is in `Examples/Hydrologic_Response_Unit/HRU_hillslope.ohq`.

4 Urban HRU

4.1 Structure

Figure 2 shows the Urban HRU (`urban_hru.json`). It has eight members:

- **Catchment** — the pervious (infiltrating) surface;
- **Impervious_Catchment** — the impervious surface, with no connection to any soil;
- **Reach**, **Impervious_Reach** — two **Trapezoidal Channel Segments** that collect the runoff of each surface: the swales, gutters, small streams and storm pipes inside the unit;
- **Soil_1**, **Soil_2**, **Soil_3** — three van Genuchten soil layers under the pervious surface;
- **Groundwater** — an **Unconfined Groundwater cell** (Section 4.7).

and six internal links: a **Catchment_link** from each surface to its reach, infiltration (**surfacewater_to_soil_link**) from the pervious surface into **Soil_1**, percolation (**soil_to_soil_link**) from **Soil_1** to **Soil_2** and from **Soil_2** to **Soil_3**, and recharge from **Soil_3** into the groundwater cell.

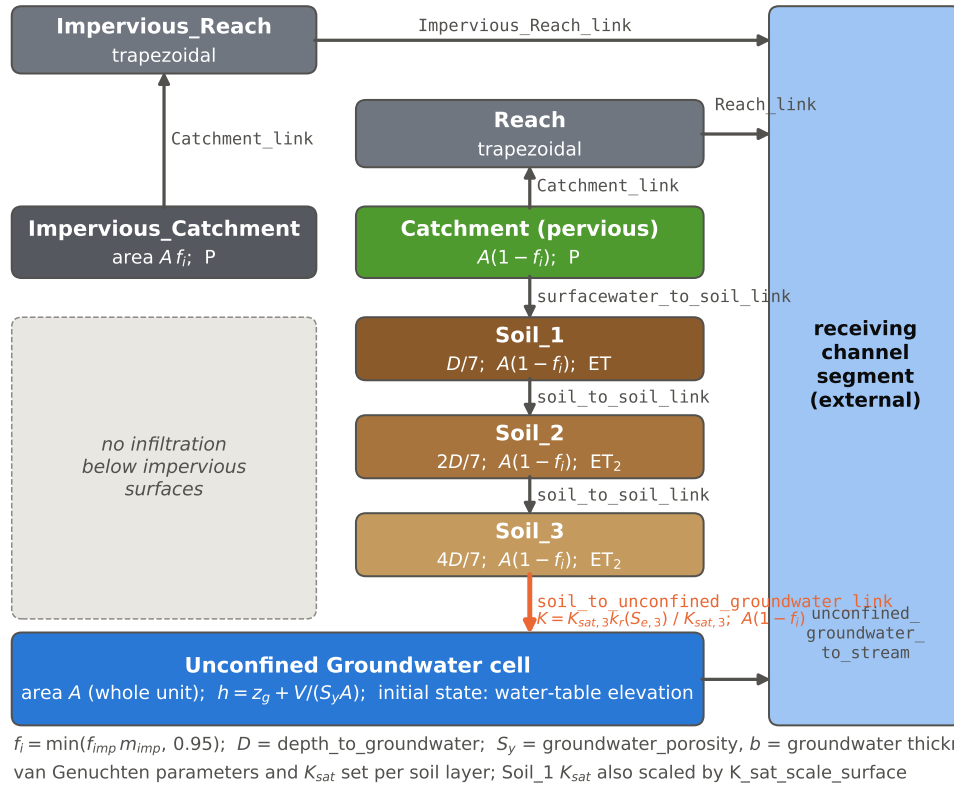


Figure 2: Urban_HRU: members (boxes, with thickness and plan area), internal links (arrows) and ports to the receiving channel segment. The recharge connector is highlighted.

4.2 Ports

A composite exposes the members that may be connected from outside as *ports*. The Urban HRU declares five, all from `unconfined_groundwater.json` or `urban_hru.json`; its baseflow leaves through `unconfined_groundwater_to_stream` (the `Linear_baseflow` connector of the HRU, Section 3, is not a port of the Urban HRU):

Port (link type)	From member	Typical target	Icon
Reach_link	Reach	the receiving channel segment	
Impervious_Reach_link	Impervious_Reach	the receiving channel segment	
unconfined_groundwater_to_stream	Groundwater	a channel segment (baseflow)	
UC_groundwater_link	Groundwater	the groundwater cell of a neighbouring Urban HRU (lateral flow)	
unconfined_groundwater_to_fixedhead	Groundwater	a fixed_head block	

Reach_link and Impervious_Reach_link are defined in urban_hru.json. Their equations are those of the channel-to-channel connector (Trapezoidal_Channel_link); they exist as separate types only so that the two outlets of the composite can be told apart when a link is drawn from it, and they have their own icons so that HRU outlets are distinguishable from links along the stream network. When a link is drawn from the unit in the interface, the port whose type matches the chosen link is used; in a script, `create link;type=Reach_link,from=SC01,to=R01` connects the Reach member of SC01.

4.3 Surface runoff and routing

Rain falls on both surfaces. Water on a surface drains to its reach with the kinematic sheet-flow relation of Catchment_link,

$$Q_s = 86400 \frac{W}{n_s} S^{1/2} (d - d_s)_+^{5/3}, \quad (4)$$

with overland width W (catchment_width), slope S (catchment_slope), Manning coefficient n_s , ponded depth d and depression storage d_s ; $(x)_+ = \max(x, 0)$. Both surfaces share W and S and have their own n_s and d_s . The reaches drain to the receiving segment through the outlet ports with Manning's equation driven by the head difference,

$$Q_r = 86400 \frac{1}{\bar{n}} A_u R^{2/3} \left(\frac{h_r - h_c}{\bar{L}} \right)^{1/2}, \quad \bar{L} = \frac{1}{2} (L_r + L_c), \quad \bar{n} = \frac{1}{2} (n_r + n_c), \quad (5)$$

where A_u is the flow area of the end with the higher head, R the hydraulic radius from the mean perimeter, and subscripts r and c refer to the internal reach and the receiving channel. The reach bottoms are placed at the surface elevation.

4.4 The impervious fraction

The impervious surface drains directly to the internal impervious reach and from there to the channel, with no chance to infiltrate. `impervious_fraction` must therefore be the *directly connected* impervious area (DCIA) — roofs, roads and parking areas whose runoff reaches a storm drain or the stream without crossing pervious ground — and not the total impervious area (TIA) mapped from land cover. The composites take it as a plain number; how it is estimated is up to the modeller. Common choices:

- **Empirical TIA–DCIA relations.** Sutherland (2000) gives DCIA as a function of TIA (both in percent, for $\text{TIA} \geq 1\%$) for five degrees of connectivity:

Drainage	DCIA
Totally connected (curb, gutter and storm sewer; roofs tied in)	TIA
Highly connected	0.4 TIA ^{1.2}
Average (mostly storm-sewered, some roofs to lawns)	0.1 TIA ^{1.5}
Somewhat disconnected	0.04 TIA ^{1.7}
Highly disconnected (swales, roofs to lawns)	0.01 TIA ²

- **Storm-drain inventories.** Where inlet locations are available, impervious surfaces within a set distance of an inlet (or of a guttered street that leads to one) can be counted as connected. Public inventories miss private systems, so this tends to be a lower bound.
- **Calibration.** The estimate is scaled by `impervious_multiplier` (Section 4.10); keeping each unit's estimate and calibrating one multiplier preserves the spatial pattern and adjusts the level.

Impervious surfaces that are *not* directly connected remain part of the pervious surface (Section 4.11).

4.5 Areas

With $f_i = \min(f_{imp} m_{imp}, 0.95)$, where f_{imp} is `impervious_fraction` and m_{imp} the `impervious_multiplier`:

Member	Plan area
Impervious_Catchment	$A f_i$
Catchment, Soil_1–Soil_3, recharge interface	$A (1 - f_i)$
Groundwater	A

The aquifer continues beneath roads and roofs, so the groundwater cell receives the whole area; it is recharged only through the pervious soil column. Were it given only the pervious area, the water table would rise $1/(1 - f_i)$ times too fast per unit of recharge — twice as fast at $f_i = 0.5$ — and any change of f_i during calibration would also change the aquifer.

4.6 Recharge through the unsaturated conductivity

`soil_to_unconfined_groundwater_link` (`unconfined_groundwater.json`) connects `Soil_3` (the source, which must be a `Soil` block) to the groundwater cell:

$$Q_r = A(1 - f_i) K_{sat,3} \frac{k_r(S_{e,3}) (h_3 - \Delta z - h_g)_+ - (h_g - h_3 + \Delta z)_+}{L}, \quad L = \frac{1}{2} (d_3 + b - \Delta z), \quad (6)$$

with d_3 the thickness of `Soil_3` and b the `aquifer_thickness` of the cell (the unit's `groundwater_thickness`), so that L runs from the centre of the soil layer to the centre of the aquifer cell, and Δz the link's `elevation_offset` (the unit's `recharge_elevation_offset`, default 0). When the soil head is higher (drainage) the conductivity is the unsaturated conductivity of the bottom layer, with the same Mualem k_r as the percolation links; when the groundwater head is higher (upward flow) the saturated conductivity is used.

The reason is the depth of urban and upland water tables. When the water table lies several metres or more below the soil column, $h_3 - h_g$ is dominated by elevation and hardly depends on how wet the soil is. With a constant conductivity the flux is then almost independent of soil moisture: in a watershed application with the water table 6–47 m below the soil and $K_{rech} = 0.1$ m/d, the constant- K link gave a potential recharge of about 150 mm/d at $S_e = 0.5$, and at $S_e = 0.2$ the suction of the bottom layer reversed it into up to 380 mm/d of upward flow

from the deep water table. Calibrating K_{rech} can only trade one error for the other. With (6) the flux falls with k_r as the bottom layer drains, and no separate recharge conductivity is needed.

Elevation offset. The soil column of a unit sits at the unit’s mean surface elevation, while the single groundwater cell has its water table near the stream: in incised terrain the two can be tens of metres apart, so that recharge falls across a large gap under a nearly constant gradient and the water table can never reach the root zone. Δz lets the recharge link evaluate `Soil_3` as if it lay Δz lower, a chosen gap G above the water table, $\Delta z = (z_{3,\text{bottom}} - h_{g,0}) - G$ (at least 0); the path length shrinks by the same amount, exactly as if the column had been moved. Nothing else moves: the soil layers, the catchments, the internal reaches and the groundwater cell keep their elevations, so surface routing is unaffected. With a small G the link reverses when `Soil_3` dries (suction larger than $G + d_3$), and evapotranspiration from the deeper layers can then draw on the groundwater.

4.7 Unconfined groundwater

The groundwater member is the stock `Unconfined Groundwater` cell of `unconfined_groundwater.json`, a cell with a free water table. Its storage V_g and piezometric head are related through the specific yield S_y (the unit’s `groundwater_porosity`, which sets both `specific_yield` and `porosity` of the cell):

$$h_g = z_g + \frac{V_g}{S_y A}, \quad V_{g,0} = S_y (h_0 - z_g) A, \quad (7)$$

where z_g is the cell’s `base_elevation`, so the water table rises by $1/S_y$ per unit depth of stored water. The initial state is the initial piezometric head h_0 (the unit’s `initial_water_table_elevation`, same datum as `surface_elevation`) and the initial storage is computed from it; the initial water table therefore does not move when S_y changes, for example during calibration. (Specifying an initial moisture content instead would tie the initial head to S_y : a calibrated value of 0.107 instead of 0.15 would raise it by 40% of the saturated thickness.) The cell reports its saturated thickness as `depth` = $h_g - z_g$. For the Urban HRU the template gained one property, `aquifer_thickness` (the unit’s `groundwater_thickness`): the thickness of the cell from its base to the bottom of the soil column. It sets the recharge path length of (6); it does not cap the storage.

Neighbouring units exchange groundwater through the stock `UC_groundwater_link`, a Darcy connector whose transmissive thickness follows the water tables:

$$Q = \frac{1}{2} (K_s + K_e) \frac{1}{2} (b_s + b_e) W \frac{h_s - h_e}{L}, \quad (8)$$

where K is the cells’ `hydraulic_conductivity` (the unit’s `groundwater_K`), b their saturated thickness (`depth`), W the interface width (`width`, the contact length of the shared boundary) and L the distance between the centroids of the two units (`length`). For example, `create link;type=UC_groundwater_link,from=SC01,to=SC02,width=3616,length=1352`. The flow is multiplied by a drainage cutoff on the cell it leaves (the upgradient cell, u):

$$f_c = \min\left(\frac{(b_u - b_{r,u})_+}{\delta}, 1\right), \quad (9)$$

with b_r the cell’s `residual_thickness` (the unit’s `groundwater_residual_thickness`, default 0) and δ the link’s `cutoff_depth` (default 0.05 m). Lateral outflow therefore tapers to zero as the cell drains to its residual thickness, and a cell cannot be emptied (or driven negative) by its neighbours. Setting b_r to the depth of the cell base below the receiving stream bed keeps lateral flow from drawing the water table below the stream: with one cell per unit there is no groundwater mound under the divide between two units, so the head difference across it is the

difference of their stream beds, and without the cutoff a high tributary unit can be drained into a lower neighbour until its stream turns losing. Exchange with the stream itself is not cut off.

Baseflow uses `unconfined_groundwater_to_stream`, the unconfined counterpart of `groundwater_to_stream`:

$$Q_b = K_b A_b \frac{h_g - h_c}{L_b}, \quad (10)$$

with h_c the head of the channel segment, A_b the bank (seepage-face) area (`area`), L_b the flow path to the stream (`length`) and K_b the link's hydraulic conductivity. Water flows back into the aquifer when the stream is higher.

4.8 Layer-specific soil

`K_sat_i`, `alpha_vG_i`, `n_vG_i`, `theta_sat_i` and `theta_res_i` ($i = 1, 2, 3$) are set for each layer, for example from soil-survey horizons averaged over the three depth intervals. The `K_sat_i` go to each layer's `K_sat_original`, so that the layer's `K_sat_scale_factor` remains free for the multiplier below.

4.9 Evapotranspiration from the deeper soil

Besides Evapotranspiration on `Soil_1`, the Urban HRU has an optional source property `Evapotranspiration_2`, applied to `Soil_2` and `Soil_3` (the root zone below the top layer). Each layer's withdrawal is scaled by its own θ/θ_s . Connecting a separate source — typically the same reference-ET series with its own `corr_fact` — lets uptake from the deeper layers be scaled independently of the top layer, which gives summer evapotranspiration a way to intercept water before it drains to the groundwater cell. Left empty, the deeper layers do not transpire.

4.10 Calibration multipliers

All three default to 1 and leave the unit unchanged at that value.

impervious_multiplier (m_{imp})

scales the impervious fraction: $f_i = \min(f_{imp} m_{imp}, 0.95)$. It is meant for calibrating the *directly connected* impervious area watershed-wide when f_{imp} comes from an empirical estimate. $m_{imp} > 1$ means that more of the land drains straight to the channel than the estimate says (for example, more roofs and driveways tied to the storm drains); $m_{imp} < 1$ means more of the impervious surface is disconnected. Area moves between the impervious catchment and the pervious catchment with its soil column; the total stays A and the groundwater cell is unaffected. Infiltration, soil storage, `Soil_1` evapotranspiration and recharge scale with the pervious area; the internal reaches and the `unconfined_groundwater_to_stream` link do not change. f_{imp} itself (for example a Sutherland estimate, Section 4.4) is fixed when the model is built; only m_{imp} is calibrated.

K_sat_scale

multiplies K_{sat} of all three layers, and therefore also the recharge conductivity of (6).

K_sat_scale_surface

multiplies K_{sat} of `Soil_1` only: its `K_sat_scale_factor` is $K_{sat_scale} \times K_{sat_scale_surface}$. It controls infiltration at the surface — reduced, for example, by the compaction of urban lawns and graded fill, which soil surveys of undisturbed profiles do not describe — without changing percolation and recharge through the deeper layers. Note that the infiltration link drives flow with the suction gradient over half the top layer, so while `Soil_1` is unsaturated its infiltration capacity is several times $K_{sat,1}$; infiltration-excess runoff from the pervious surface appears only when rain intensity exceeds that capacity.

Bind a model parameter to them at the composite level,

```
create parameter; type=Parameter, name=p_imp, low=0.3, high=3, value=1, prior_distribution=log-normal
setasparameter; object=SC01, quantity=impervious_multiplier, parametername=p_imp
```

and the value reaches the members through the composite's mappings. When a parameter changes, the member areas are re-evaluated and the initial soil storages $\theta_0 A_{\ell d \ell}$ recomputed, so the initial moisture content stays θ_0 ; the code generator (`ohq_generate`) compiles the same mappings and initial storages as expressions of the parameter vector.

4.11 Disconnected impervious area

Only the *connected* impervious fraction is an impervious surface. Impervious surfaces that drain onto lawns — roofs with leaders to splash blocks, driveways sloping to grass — belong to the pervious catchment $A(1 - f_i)$ and are treated as soil: rain on them can infiltrate, and the Soil_1 evapotranspiration and the recharge interface act over them. Neither the concentration of their runoff onto the adjacent lawn nor the absence of transpiration from roofs is represented. If that matters, a third Catchment for the disconnected area, joined to Catchment by a Catchment_link, would represent both; it is not part of the composite.

5 Using the units

5.1 Templates

The HRU lists `main_components.json`, `unsaturated_soil.json` and `groundwater.json` in its `requires`; the Urban HRU lists `main_components.json`, `unsaturated_soil.json`, `unconfined_groundwater.json` and `open_channel.json` (not `groundwater.json`), so selecting either alone in the interface is enough. The evapotranspiration source is not a member; load `soil_evapotranspiration_models.json` for the Evapotranspiration_Time_Series (Soil) source. In a script, add each template explicitly.

5.2 A worked example

Examples/Urban_HRU/Urban_HRU_example.ohq is one 50 ha unit with 25 % impervious area, draining to an 800 m trapezoidal stream that ends at a fixed head, forced by 30 days of hourly rain and daily reference ET:

```
loadtemplate; filename = main_components.json
addtemplate; filename = unsaturated_soil.json
addtemplate; filename = unconfined_groundwater.json
addtemplate; filename = open_channel.json
addtemplate; filename = soil_evapotranspiration_models.json
addtemplate; filename = urban_hru.json
create source; type=Precipitation, name=Rain, timeseries=precipitation.csv
create source; type=Evapotranspiration_Time_Series (Soil), name=ET, corr_fact=0.8, ET_timeseries=
  evapotranspiration.csv
create composite; type=Urban_HRU, name=Neighbourhood, area=500000, impervious_fraction=0.25,
  surface_elevation=100, depth_to_groundwater=3.5, groundwater_thickness=12,
  Precipitation=Rain, Evapotranspiration=ET, catchment_slope=0.03, catchment_width=2500,
  K_sat_1=0.6, K_sat_2=0.4, K_sat_3=0.8, alpha_vG_1=1.5, ...,
  groundwater_porosity=0.15, initial_water_table_elevation=88.5,
  reach_length=400, impervious_reach_length=400, ...
create block; type=Trapezoidal Channel Segment, name=Stream, base_width=3, side_slope=2,
  ManningCoeff=0.035, bottom_elevation=88, length=800, depth=0.1
create block; type=fixed_head, name=Outlet, head=87
create link; type=Reach_link, from=Neighbourhood, to=Stream, name=Pervious_outlet
create link; type=Impervious_Reach_link, from=Neighbourhood, to=Stream, name=Impervious_outlet
create link; type=unconfined_groundwater_to_stream, from=Neighbourhood, to=Stream, name=Baseflow,
  area=8000, length=150, hydraulic_conductivity=5
```

```
create link;type=channel2fixed,from=Stream,to=Outlet,name=Stream_outlet
```

(Line breaks inside `create` commands are for display only.) Members are addressed as `Neighbourhood_Soil_3`, `Neighbourhood_recharge` and so on in outputs and scripts.

Figure 3 and Table 2 show the run. All impervious runoff reaches the stream within hours of each storm, while the pervious surface produces almost none because the rain rate rarely exceeds the infiltration capacity. About three quarters of the rain infiltrates; recharge starts at 8 mm/d as the initially wet bottom layer drains, falls with its unsaturated conductivity and settles near the rate at which water percolates through the column, and the water table rises gradually and feeds baseflow between storms.

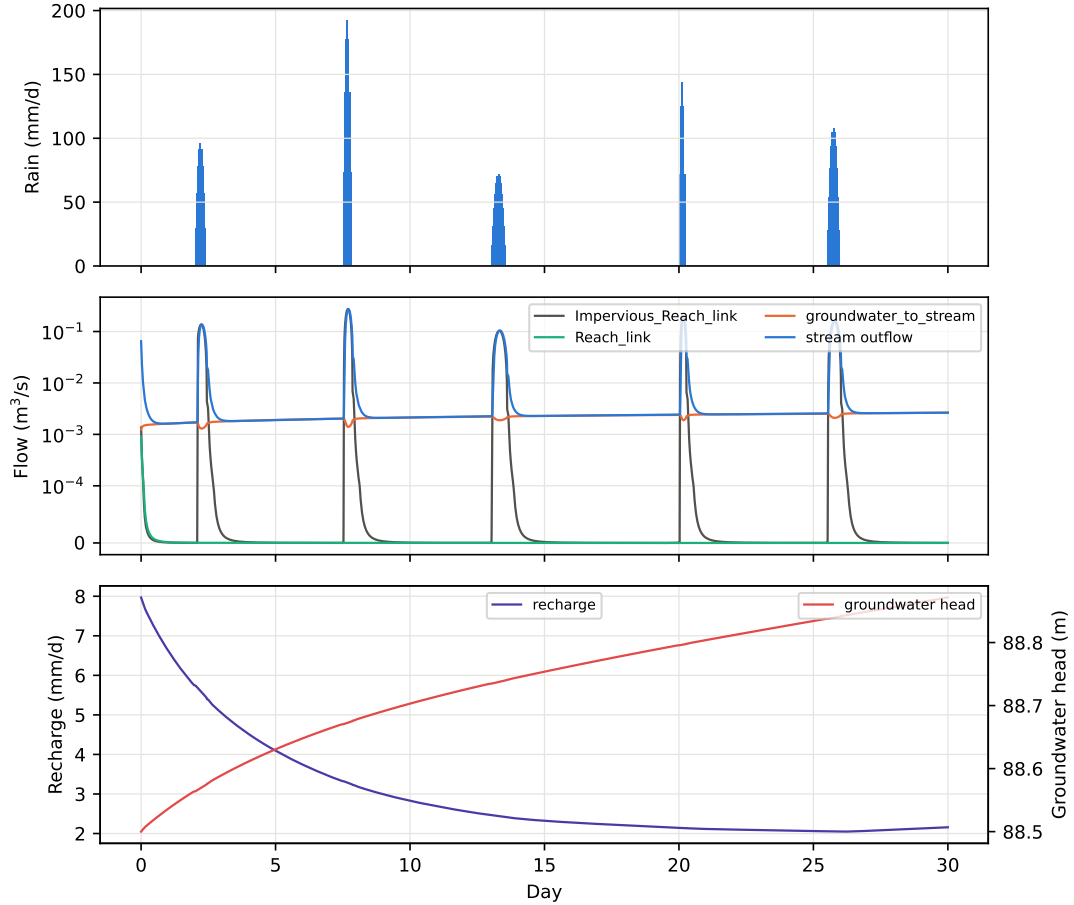


Figure 3: The worked example: rain, outflows of the three ports and of the stream, recharge and groundwater head.

Volume over 30 days	m ³
Rain	74,336
Impervious outlet (<code>Impervious_Reach_link</code>)	18,370
Pervious outlet (<code>Reach_link</code>)	4
Infiltration	55,749
Recharge	33,526
Baseflow (<code>groundwater_to_stream</code>)	5,717
Stream outflow	24,179
Groundwater head rise (m)	0.37

Table 2: Water balance of the worked example.

6 Property reference

Generated from the two resource files by `docs/make_hru_figures.py`. “In” gives the units that expose the property; “Applied to” lists the members that receive it; where the defaults differ between the units they are separated by a slash.

Property	Unit	Default	In	Applied to
<code>area</code>	m ²	1000	both	member areas
<code>recharge_K</code>	m/day	1	HRU	recharge
<code>surface_elevation</code>	m	0	both	Catchment, Groundwater, Impervious_Catchment, Impervious_Reach, Reach, Soil_1, Soil_2, Soil_3
<code>depth_to_groundwater</code>	m	3.5	both	Soil_1, Soil_2, Soil_3
<code>groundwater_thickness</code>	m	5	both	Groundwater
<code>Precipitation</code>			both	Catchment, Impervious_Catchment
<code>Evapotranspiration</code>			both	Soil_1
<code>catchment_slope</code>		0.01	both	Catchment, Impervious_Catchment
<code>catchment_width</code>	m	30	both	Catchment, Impervious_Catchment
<code>manning_coefficient</code>		0.05	both	Catchment
<code>depression_storage</code>	m	0	both	Catchment
<code>runoff_coefficient</code>		1	both	Catchment, Impervious_Catchment
<code>K_sat</code>	m/day	1	HRU	Soil_1, Soil_2, Soil_3
<code>alpha_vG</code>	1/m	1	HRU	Soil_1, Soil_2, Soil_3
<code>n_vG</code>		1.41	HRU	Soil_1, Soil_2, Soil_3
<code>theta_sat</code>		0.4	HRU	Soil_1, Soil_2, Soil_3
<code>theta_res</code>		0.05	HRU	Soil_1, Soil_2, Soil_3
<code>soil_specific_storage</code>	1/m	0.01	both	Soil_1, Soil_2, Soil_3
<code>initial_moisture_content</code>		0.2	both	Soil_1, Soil_2, Soil_3
<code>groundwater_K</code>	m/day	1	both	Groundwater
<code>groundwater_porosity</code>		0.2/0.4	both	Groundwater
<code>groundwater_specific_storage</code>	1/m	0.01	HRU	Groundwater
<code>initial_groundwater_moisture_content</code>		0.25	HRU	Groundwater
<code>impervious_fraction</code>		0.2	Urban	member areas
<code>groundwater_residual_thickness</code>	m	0	Urban	Groundwater
<code>recharge_elevation_offset</code>	m	0	Urban	recharge
<code>Evapotranspiration_2</code>			Urban	Soil_2, Soil_3
<code>K_sat_1</code>	m/day	1	Urban	Soil_1
<code>K_sat_2</code>	m/day	1	Urban	Soil_2
<code>K_sat_3</code>	m/day	1	Urban	Soil_3
<code>alpha_vG_1</code>	1/m	1	Urban	Soil_1
<code>alpha_vG_2</code>	1/m	1	Urban	Soil_2
<code>alpha_vG_3</code>	1/m	1	Urban	Soil_3
<code>n_vG_1</code>		1.41	Urban	Soil_1
<code>n_vG_2</code>		1.41	Urban	Soil_2
<code>n_vG_3</code>		1.41	Urban	Soil_3
<code>theta_sat_1</code>		0.4	Urban	Soil_1
<code>theta_sat_2</code>		0.4	Urban	Soil_2
<code>theta_sat_3</code>		0.4	Urban	Soil_3
<code>theta_res_1</code>		0.05	Urban	Soil_1

Property	Unit	Default	In	Applied to
theta_res_2		0.05	Urban	Soil_2
theta_res_3		0.05	Urban	Soil_3
initial_water_table_elevation	m	93	Urban	Groundwater
impervious_manning_coefficient		0.015	Urban	Impervious_Catchment
impervious_depression_storage	m	0.001	Urban	Impervious_Catchment
reach_base_width	m	1	Urban	Reach
reach_side_slope		2	Urban	Reach
reach_manning_coefficient		0.035	Urban	Reach
reach_length	m	10	Urban	Reach
initial_reach_depth	m	0.01	Urban	Reach
impervious_reach_base_width	m	1	Urban	Impervious_Reach
impervious_reach_side_slope		2	Urban	Impervious_Reach
impervious_reach_manning_coefficient		0.02	Urban	Impervious_Reach
impervious_reach_length	m	10	Urban	Impervious_Reach
initial_impervious_reach_depth	m	0.01	Urban	Impervious_Reach
impervious_multiplier		1	Urban	member areas
K_sat_scale		1	Urban	Soil_1, Soil_2, Soil_3
K_sat_scale_surface		1	Urban	

7 Limitations and pitfalls

- **Fixed layer split.** The three soil layers are always $D/7$, $2D/7$ and $4D/7$; D controls the storage, the recharge path length and the heads together, so calibrating it moves all three at once.
- **Deep water tables in the HRU.** When the water table is far below the soil column, the HRU's constant recharge conductivity drains the column regardless of its wetness and pulls water upward when the bottom layer dries (Section 4.6); keep `recharge_K` small or use the Urban HRU, whose recharge follows the unsaturated conductivity.
- **Confined-style defaults in the HRU.** With $S_s = 0.01 \text{ m}^{-1}$ the head responds metres to small changes of storage. For an unconfined aquifer set $S_s = \phi/b$.
- **Outlet link length.** (5) averages the length of the internal reach with that of the receiving segment, so a long receiving segment lowers the outlet slope.
- **Internal reaches are generic.** Their shape and roughness are properties of the unit, not derived from its drainage network; they mostly control the timing of the runoff peak.
- **Evapotranspiration.** The HRU transpires from `Soil_1` only; the Urban HRU from `Soil_1` and, with `Evapotranspiration_2`, from `Soil_2` and `Soil_3`. Each layer's rate is scaled by its own θ/θ_s . The groundwater cell does not transpire.
- **Hourly rainfall and infiltration excess.** With hourly forcing, short bursts of high intensity are averaged out, so infiltration-excess runoff from pervious surfaces is underestimated; `K_sat_scale_surface` can compensate in calibration.
- **Disconnected impervious area** is treated as pervious soil (Section 4.11).

- **One unit, one soil column.** Spatial variation inside a sub-catchment is represented only through its areas; use more units for more variation.

Reference

Sutherland, R. C. (2000). Methods for estimating the effective impervious area of urban watersheds. *Watershed Protection Techniques* 2(4), 282–284.